Aggregate Demand and Aggregate Supply Analysis

ECON 101 - Macroeconomics

Muhebullah Karimzada

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Chapter Outline

- **9.1** Aggregate Demand
- **9.2** Aggregate Supply
- **9.3** Macroeconomic Equilibrium in the Long Run and the Short Run

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9.1 Learning Objective

Goal

Identify the determinants of aggregate demand and distinguish between a movement along the AD curve and a shift of the curve.

- Real GDP has four components:
  - Consumption (C)
  - Investment (I)
  - Government purchases (G)
  - Net exports (NX)
- Adding these gives:

$$Y = C + I + G + NX.$$

- In this chapter we treat each component as a **function of the price level and other variables** and use them to build the **aggregate demand curve**.

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Aggregate Demand and Aggregate Supply Model

- We have already modeled:
  - long-run economic growth, and
  - short-run output determination using the AE model.
- Now we extend our short-run model to understand:
  - why real GDP, employment, and the price level fluctuate together.
- **Aggregate demand and aggregate supply model:**
  - explains short-run fluctuations in real GDP and the price level.

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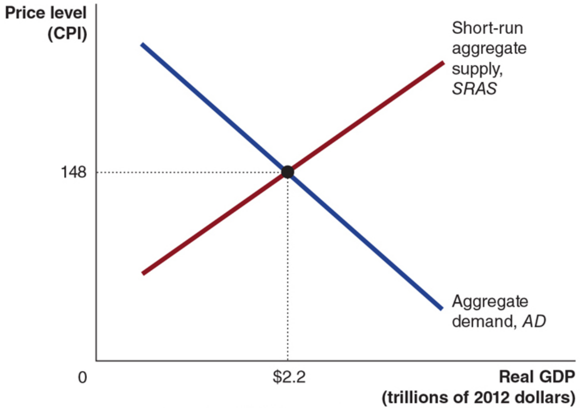
Figure 9.1 – Aggregate Demand and Aggregate Supply

Figure 9.1 Aggregate Demand and Aggregate Supply

In the short run, real GDP and the price level are determined by the intersection of the aggregate demand curve and the short-run aggregate supply curve. In the figure, real GDP is measured on the horizontal axis, and the price level is measured on the vertical axis by the CPI. In this example, the equilibrium real GDP is $1.8 trillion and the equilibrium price level is 125.



AD and SRAS: Basic Definitions

- **Aggregate demand (AD) curve:**
  - Shows the relationship between the price level and the quantity of real GDP demanded by:
    - households, firms, the government, and the foreign sector.
- **Short-run aggregate supply (SRAS) curve:**
  - Shows the relationship in the short run between the price level and the quantity of real GDP supplied by firms.
- The intersection of AD and SRAS determines the **short-run macroeconomic equilibrium**.

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Why the Aggregate Demand Curve Slopes Down

- As the **price level** rises:
  - households feel poorer,
  - interest rates tend to rise,
  - domestic goods become more expensive relative to foreign goods.
- Each of these channels reduces one or more components of real GDP:
  - C (consumption),
  - I (investment),
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- Result: a **higher** price level is associated with a **smaller** quantity of real GDP demanded.

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The Wealth Effect: Price Level and Consumption

- Household consumption is mostly determined by income, but also by **wealth**.
- Some household wealth is held in **nominal assets** (cash, bank deposits, bonds).
- When the price level rises:
  - the **real value** of these nominal assets falls,
  - households feel poorer and reduce consumption.
- Implication:
  - Higher price level $\Rightarrow$ lower consumption.

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- When the price level rises:
  - households and firms need more money to buy and sell goods and services.
- They may:
  - borrow more,
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  - sell financial assets such as bonds.
- This extra demand for money pushes up the **interest rate**:
  - Higher interest rates discourage borrowing,
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The International-Trade Effect: Price Level and Net Exports

- When Canadian price levels rise:
  - Canadian goods become more expensive relative to foreign goods.
- Consequences:
  - Exports fall (foreigners buy fewer Canadian goods),
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- Net exports $NX = X - M$ therefore fall.
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Summary: Why AD Slopes Down

- Each effect links a higher price level to lower spending:
  - Wealth effect $\Rightarrow$ lower C,
  - Interest-rate effect $\Rightarrow$ lower I,
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- Together, these effects imply:
  - A downward-sloping **aggregate demand curve**.

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Movements Along vs. Shifts of the AD Curve

- The AD curve shows the relationship between the **price level** and real GDP demanded, **holding everything else constant**.
- **Movement along AD:**
  - Caused by a change in the price level itself,
  - not by a change in any component of spending.
- **Shift of AD:**
  - Caused by a change in C, I, G, or NX from any source *other than* the price level.

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  - Caused by a change in C, I, G, or NX from any source *other than* the price level.

Table 9.1 – Variables That Shift the AD Curve (1 of 4)

Government Policy: Monetary Policy

- **Monetary policy:**
  - Actions the central bank (e.g., the Bank of Canada) takes to manage the money supply and interest rates to pursue macroeconomic objectives.

Change	**Effect on AD**
Increase in interest rates	Reduces investment and some consumption; **AD shifts left**.
Decrease in interest rates	Increases investment and some consumption; **AD shifts right**.

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Table 9.1 – Variables That Shift the AD Curve (2 of 4)

Government Policy: Fiscal Policy

- **Fiscal policy:**
  - Changes in federal taxes and purchases intended to achieve macroeconomic objectives.

Change	Effect on AD
Increase in personal income taxes	Reduces disposable income and consumption; **AD shifts left**.
Decrease in personal income taxes	Raises disposable income and consumption; **AD shifts right**.
Increase in government purchases	Directly raises G; **AD shifts right**.
Decrease in government purchases	Directly lowers G; **AD shifts left**.
Increase in business taxes	Reduces investment; **AD shifts left**.
Decrease in business taxes	Increases investment; **AD shifts right**.

Table 9.1 – Variables That Shift the AD Curve (2 of 4)

Government Policy: Fiscal Policy

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  - Changes in federal taxes and purchases intended to achieve macroeconomic objectives.

Change	**Effect on AD**
Increase in personal income taxes	Reduces disposable income and consumption; **AD shifts left**.
Decrease in personal income taxes	Raises disposable income and consumption; **AD shifts right**.
Increase in government purchases	Directly raises G; **AD shifts right**.
Decrease in government purchases	Directly lowers G; **AD shifts left**.
Increase in business taxes	Reduces investment; **AD shifts left**.
Decrease in business taxes	Increases investment; **AD shifts right**.

Table 9.1 – Variables That Shift the AD Curve (3 of 4)

Changes in Expectations

- Households and firms may become more optimistic or pessimistic about the future.

Change	Effect on AD
Households expect higher future incomes	Increases current consumption; **AD shifts right**.
Households expect lower future incomes	Decreases current consumption; **AD shifts left**.
Firms expect higher future profitability	Increases investment; **AD shifts right**.
Firms expect lower future profitability	Reduces investment; **AD shifts left**.

Table 9.1 – Variables That Shift the AD Curve (4 of 4)

Changes in Foreign Variables

Change	**Effect on AD**
Domestic GDP growth rate rises relative to other countries	Imports rise faster than exports; net exports fall; **AD shifts left**.
Domestic GDP growth rate falls relative to other countries	Imports grow slowly, exports may rise; net exports increase; **AD shifts right**.
Increase in the value of the Canadian dollar (appreciation)	Canadian exports become more expensive; imports cheaper; net exports fall; **AD shifts left**.
Decrease in the value of the Canadian dollar (depreciation)	Canadian exports cheaper; imports more expensive; net exports rise; **AD shifts right**.

9.2 Learning Objective

Goal

Identify the determinants of aggregate supply and distinguish between a movement along the SRAS curve and a shift of the curve.

- **Aggregate supply** refers to the quantity of goods and services that firms are willing and able to supply.
- The relationship between this quantity and the price level differs:
  - in the **long run**, and
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Long-Run Aggregate Supply (LRAS)

- **Long-run aggregate supply (LRAS) curve:**
  - Shows the relationship in the long run between the price level and the quantity of real GDP supplied.
- In the long run, real GDP is determined by:
  - number of workers,
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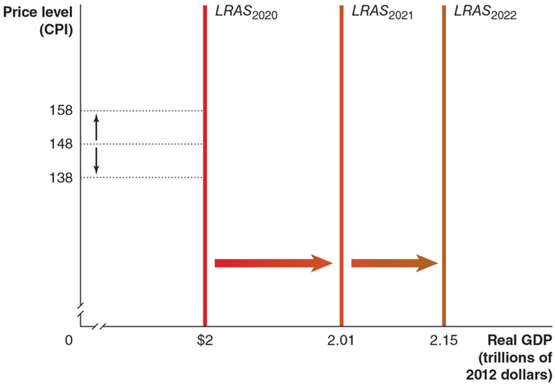
Figure 9.2 – Long-Run Aggregate Supply Curve

Figure 9.2 The Long-Run Aggregate Supply Curve

Changes in the price level do not affect the level of aggregate supply in the long run. Therefore, the long-run aggregate supply (*LRAS*) curve is a vertical line at potential GDP. For instance, the price level was 138 in 2020 and potential GDP was $2 trillion. If the price level had been 148, or if it had been 125, long-run aggregate supply would still have been a constant $2 trillion. Each year, the long-run aggregate supply curve shifts to the right as the number of workers in the economy increases, more machinery and equipment are accumulated, and technological change occurs.



Short-Run Aggregate Supply (SRAS)

- **Short-run aggregate supply (SRAS) curve:**
  - Shows the relationship in the short run between the price level and the quantity of real GDP supplied by firms.
- Unlike LRAS, the SRAS curve is **upward sloping**.
- Why?
  - As prices of final goods and services rise,
  - prices of inputs (wages, raw materials) rise more slowly,
  - so profits increase and firms are willing to produce more.
- Also, some firms are slow to adjust their prices when the price level changes.

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Why Is SRAS Upward Sloping?

- Economists believe some firms and workers **fail to predict** changes in the price level accurately.
- Three main explanations for an upward-sloping SRAS:
  1. Contracts make some wages and prices **sticky**.
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Sticky Wages and Prices

- Wages and prices are **sticky** if they do not respond quickly to changes in demand or supply.
- Many wages are set in long-term contracts:
  - If the price level rises unexpectedly, real wages temporarily fall,
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- Similarly, some output prices are fixed by contracts or norms:
  - Firms may delay price changes, so higher demand translates into higher output in the short run.

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- Firms are often slow to adjust wages, even outside formal contracts:
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  - Firms dislike cutting nominal wages because it is bad for morale.
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- **Menu costs:**
  - Costs of changing prices (e.g., printing new catalogues, updating price lists).
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Movements Along vs. Shifts of the SRAS Curve

- The SRAS curve shows the relationship between the price level and real GDP supplied, **holding other variables constant**.
- Movement along SRAS:
  - Caused by a change in the price level itself.
- Shift of SRAS:
  - Caused by changes in:
    - labour and capital,
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Figure 9.3 How Expectations of the Future Price Level Affect the Short-Run Aggregate Supply Curve

The *SRAS* curve shifts to reflect workers' and firms' expectations of future prices.

1. If workers and firms expect that the price level will rise by 7 percent, from 138 to 148, they will adjust their wages and prices by that amount.
2. Holding constant all other variables that affect aggregate supply, the short-run aggregate supply curve will shift to the left.

If workers and firms expect that the price level will be lower in the future, the short-run aggregate supply curve will shift to the right.

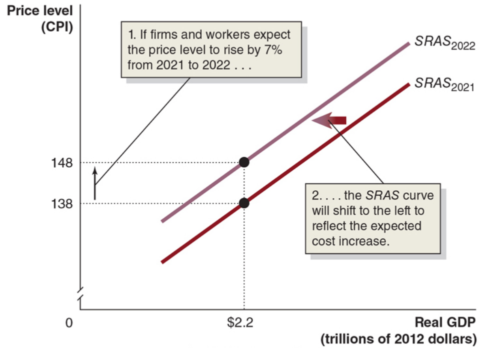

Table 9.2 – Variables That Shift SRAS (1 of 3)

Changes in Labour, Capital, and Technology

Change	**Effect on SRAS**
Increase in the labour force or capital stock	More production possible at any price level; **SRAS shifts right**.
Decrease in the labour force or capital stock	Less production possible at any price level; **SRAS shifts left**.
Improvements in technology	Higher productivity; more output at any price level; **SRAS shifts right**.

Changes in Expectations about the Price Level

Change	Effect on SRAS
Workers and firms expect a higher future price level	They set higher wages and prices now; **SRAS shifts left**.
Workers and firms realize they previously underestimated the price level	They adjust wages and prices upward; **SRAS shifts left**.
Workers and firms expect a lower future price level	They accept lower wages and prices; **SRAS shifts right**.

Supply Shocks

- **Supply shock:**
  - Unexpected event that causes SRAS to shift.

Shock	Effect on SRAS
Unexpected increase in the price of an important input (e.g., oil)	Raises production costs; **SRAS shifts left**.
Unexpected decrease in the price of an important input	Lowers production costs; **SRAS shifts right**.
Pandemic shutdowns that close businesses	Reduce available supply; **SRAS shifts left**.

Table 9.2 – Variables That Shift SRAS (3 of 3)

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  - Unexpected event that causes SRAS to shift.

Shock	Effect on SRAS
Unexpected increase in the price of an important input (e.g., oil)	Raises production costs; **SRAS shifts left**.
Unexpected decrease in the price of an important input	Lowers production costs; **SRAS shifts right**.
Pandemic shutdowns that close businesses	Reduce available supply; **SRAS shifts left**.

9.3 Learning Objective

Goal

Use the AD–AS model to illustrate the difference between short-run and long-run macroeconomic equilibrium.

- We now bring together:
  - the aggregate demand curve,
  - the short-run aggregate supply curve,
  - and the long-run aggregate supply curve.

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Long-Run Macroeconomic Equilibrium

- **Long-run macroeconomic equilibrium** occurs when:
  - AD and SRAS intersect at the LRAS level of output.
- That is:
  - real GDP equals potential (full-employment) GDP, and
  - firms and workers have correctly anticipated the price level.

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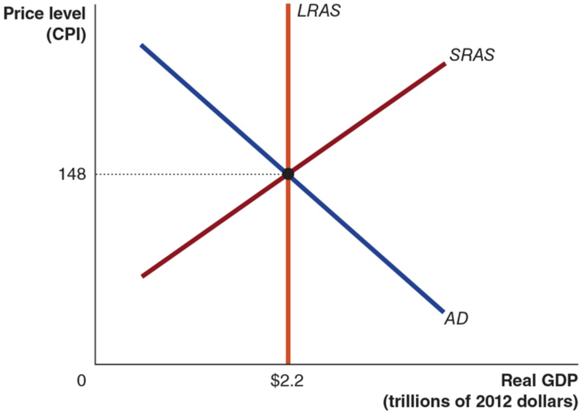
Figure 9.4 – Long-Run Macroeconomic Equilibrium

Figure 9.4 Long-Run Macroeconomic Equilibrium

In long-run macroeconomic equilibrium, the *AD* and *SRAS* curves intersect at a point on the *LRAS* curve. In this case, equilibrium occurs at real GDP of $2.2 trillion and a price level of 148.



Why Equilibrium Occurs at Potential GDP in the Long Run

- Suppose the economy starts at long-run equilibrium:
  - price level $P = 148$,
  - real GDP $= \$2.2$ trillion,
  - expected future price level $= 148$,
  - LRAS fixed at $2.2 trillion.
- We will see how various shocks move the economy away from this point in the short run, and how the economy returns to potential GDP in the long run as wages and prices adjust.

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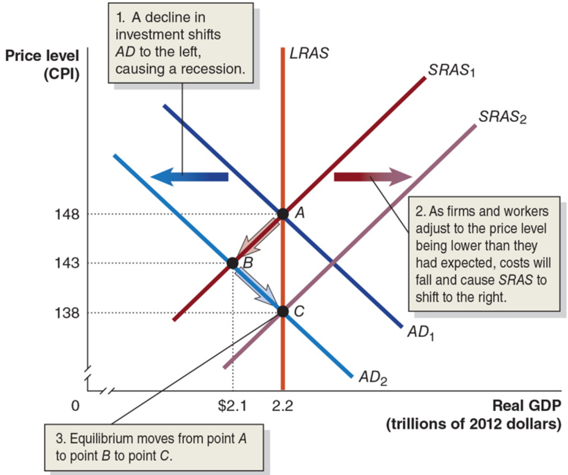
Figure 9.5 – Decrease in AD: Short Run

Figure 9.5 The Short-Run and Long-Run Effects of a Decrease in Aggregate Demand

In the short run, a decrease in aggregate demand causes a recession. In the long run, it causes only a decrease in the price level.



Short-Run Effects of a Decrease in Aggregate Demand

- Suppose interest rates rise (e.g., due to tighter monetary policy).
- Consequences:
  - Firms and households reduce planned investment.
  - Aggregate demand shifts left (from AD_1 to AD_2).
- Short run:
  - real GDP falls below potential,
  - unemployment rises,
  - the price level falls from 148 to a lower value.

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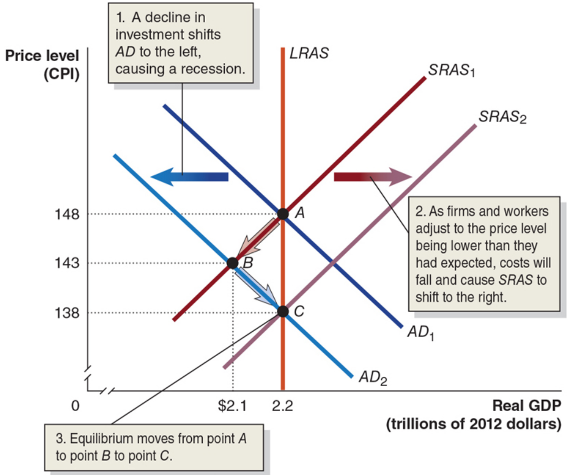

Long-Run Adjustment after a Decrease in AD

- With output below potential:
  - unemployment is higher than the natural rate,
  - workers are willing to accept lower wages,
  - firms anticipate lower input prices.
- Over time:
  - nominal wages and other input prices fall,
  - SRAS shifts to the right,
  - real GDP rises back to potential.
- Long-run result:
  - the economy returns to potential GDP,
  - but the price level is lower than initially.

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Figure 9.6 – Increase in AD: Short Run

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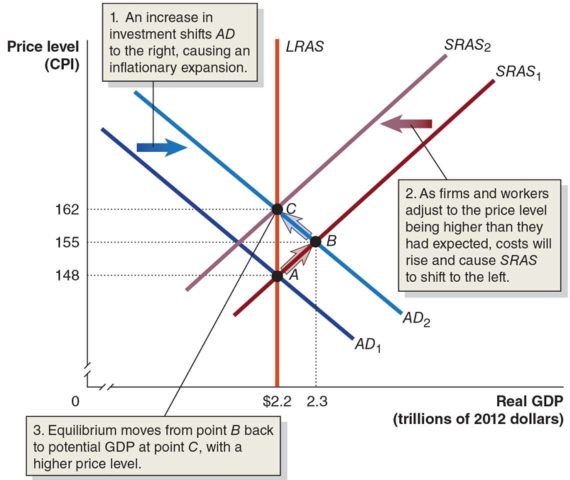
In the short run, an increase in aggregate demand causes an increase in real GDP. In the long run, it causes only an increase in the price level.



Short-Run Effects of an Increase in Aggregate Demand

- Suppose firms become more optimistic about the future.
- They increase their investment, shifting AD to the right (from AD_1 to AD_2).
- Short run:
  - real GDP rises above potential,
  - unemployment falls below the natural rate,
  - the price level rises.

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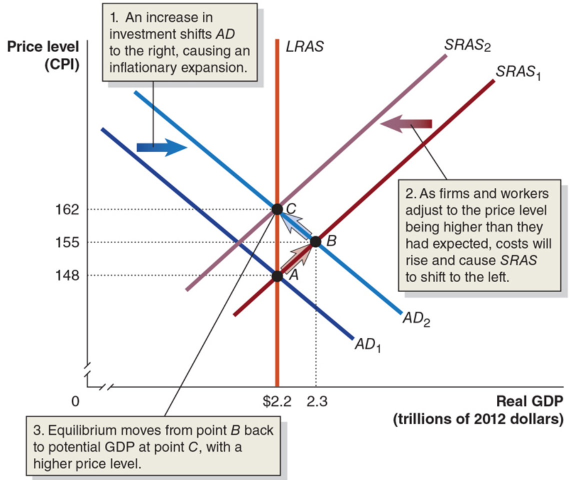

Long-Run Adjustment after an Increase in AD

- With output above potential:
  - labour markets are tight,
  - firms bid up wages to attract workers,
  - higher demand for goods and services supports higher prices.
- Over time:
  - expectations of the price level rise,
  - wages and input prices increase,
  - SRAS shifts left.
- Long-run result:
  - the economy returns to potential GDP,
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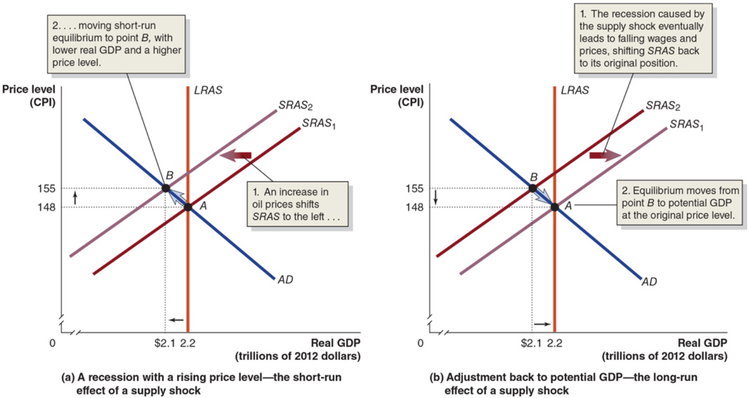
Figure 9.7 – Supply Shock: Short Run

Figure 9.7 The Short-Run and Long-Run Effects of a Supply Shock

Panel (a) shows that a supply shock, such as a large increase in oil prices, will cause a recession and a higher price level in the short run. The recession caused by the supply shock increases unemployment and reduces output. In panel (b), rising unemployment and falling output result in workers being willing to accept lower wages and firms being willing to accept lower prices. The short-run aggregate supply curve shifts from $SRAS_2$ to $SRAS_1$. Equilibrium moves from point *B* back to potential GDP and the original price level at point *A*.



Negative Supply Shock and Stagflation

- **Supply shock:**
  - Unexpected event that causes SRAS to shift.
- Example: a sudden increase in oil prices.
- Short-run effects:
  - SRAS shifts left,
  - real GDP falls,
  - unemployment rises,
  - price level rises.
- This combination of **inflation** and **recession** is called **stagflation**.

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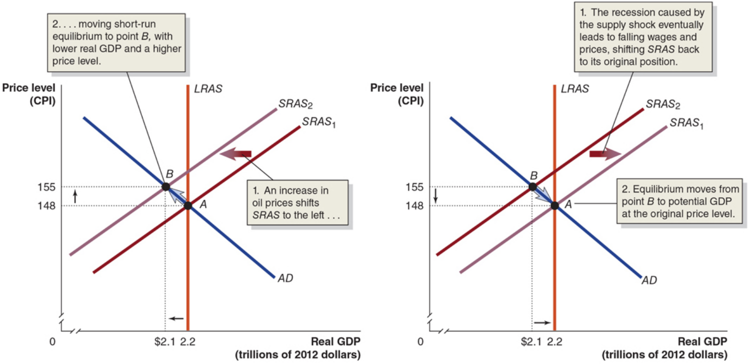

(a) A recession with a rising price level—the short-run effect of a supply shock

(b) Adjustment back to potential GDP—the long-run effect of a supply shock

Long-Run Adjustment after a Supply Shock

- With output below potential and high unemployment:
  - workers are willing to accept lower wages,
  - firms reduce prices to clear inventories.
- As expectations of future prices fall:
  - SRAS shifts right again.
- Eventually:
  - the economy returns to potential GDP,
  - and the price level may fall back toward its initial value.

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- Eventually:
  - the economy returns to potential GDP,
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Long-Run Adjustment after a Supply Shock

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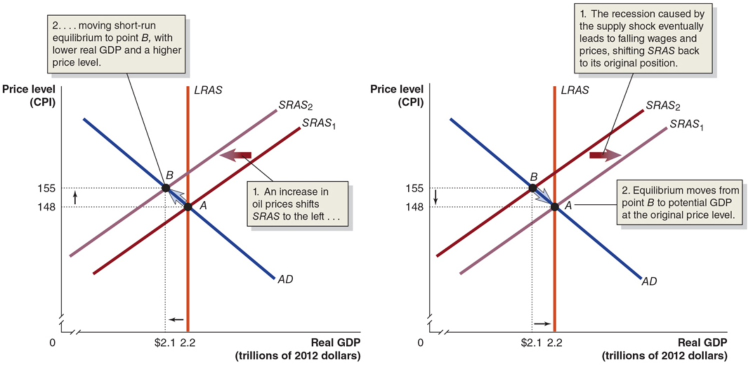
Figure 9.7 The Short-Run and Long-Run Effects of a Supply Shock

Panel (a) shows that a supply shock, such as a large increase in oil prices, will cause a recession and a higher price level in the short run. The recession caused by the supply shock increases unemployment and reduces output. In panel (b), rising unemployment and falling output result in workers being willing to accept lower wages and firms being willing to accept lower prices. The short-run aggregate supply curve shifts from $SRAS_2$ to $SRAS_1$. Equilibrium moves from point *B* back to potential GDP and the original price level at point *A*.



(a) A recession with a rising price level—the short-run effect of a supply shock

(b) Adjustment back to potential GDP—the long-run effect of a supply shock

Policy Trade-Offs after a Supply Shock

- Left alone, the economy eventually returns to full employment, but:
  - it may take several years,
  - unemployment can remain high for a long time.
- Policymakers may choose to:
  - use fiscal or monetary policy to increase AD,
  - reducing the depth and length of the recession,
  - but at the cost of permanently higher price levels.

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- Both supply and demand shocks can cause recessions.
- Question: *Which type of shock played a larger role in the 2020 COVID-19 recession?*
- A **negative supply shock**:
  - closes businesses,
  - reduces productive capacity,
  - shifts SRAS left,
  - reduces output and raises prices.

Apply the Concept: COVID-19 – Supply Shock or Demand Shock? (1 of 3)

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Apply the Concept: COVID-19 – Supply vs. Demand (2 of 3)

- A **negative demand shock**:
  - reduces household and business demand for goods and services,
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- In the first half of 2020:
  - Real GDP fell from about $2.12 trillion at the end of 2019
  - to about $1.85 trillion by mid-2020.
- Consumer prices:
  - The seasonally adjusted CPI was 137.3 in January 2020.
  - It was 136.7 in July 2020, when output had started to recover.
- Prices **fell** during the recession:
  - This pattern is more consistent with a **demand shock** than with a pure supply shock.
  - Conclusion: a fall in aggregate demand played a larger role in the 2020 recession, though supply disruptions also mattered.

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Summary

- The **aggregate demand curve** slopes downward because:
  - a higher price level reduces consumption (wealth effect),
  - reduces investment (interest-rate effect),
  - and reduces net exports (international-trade effect).
- Changes in government policy, expectations, and foreign variables shift AD.
- The **long-run aggregate supply** curve is vertical at potential GDP; the **short-run aggregate supply** curve is upward sloping due to sticky wages and prices.
- Long-run equilibrium occurs where AD and SRAS intersect LRAS.
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